

Interpopulation variability between populations of *Gammarus fossarum*: predation risk, fear and anxiety

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BACKGROUND

Ecology of fear considers **predator-induced stresses** as important drivers of **demographic strategies, behavioral** and **physiological** changes.

- **Fear**: an **immediate** consequence of a stress allowing to face adequately an **identified** danger;
- **Anxiety**: a consequence of a stress occurring when the threat is **not clearly identified** and which is generally **sustained** in absence of stimuli or in a new environment.

OBJECTIVES

- To document the effect of **local predation risk** on **antipredator behaviours**.
- To assess **flexibility** of antipredator behaviours according to a **current predation risk**.

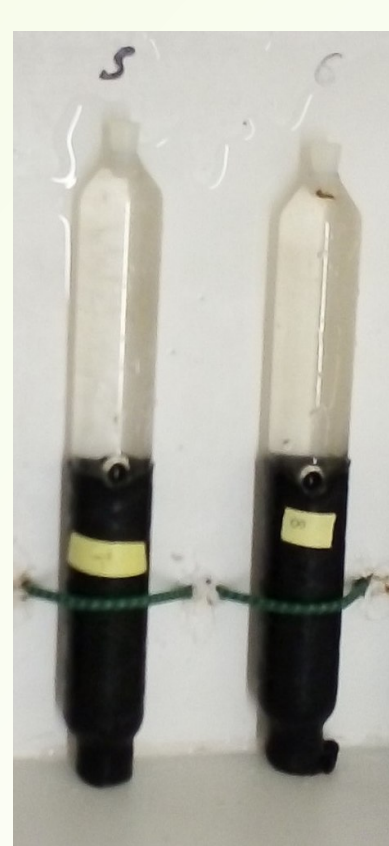
**Predation risk proxy:
Fish biomass**

METHODS AND RESULTS

Antipredator behaviours



Refuge use



Phototaxis

15 gammarid populations
80 ind/pop/behaviour

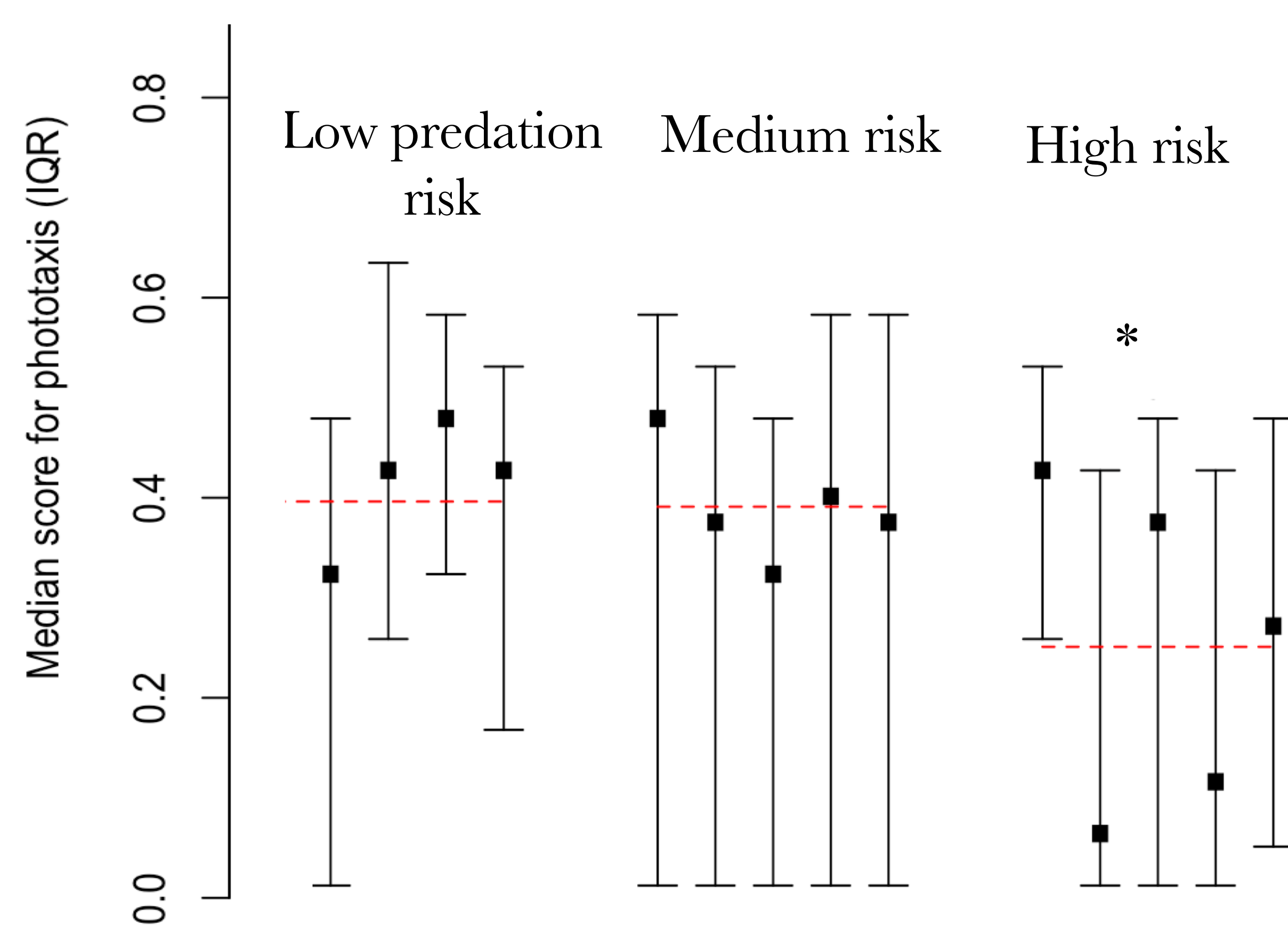


Fig1. Phototaxis score according to predation risk.
1: always in the light, 0: always in obscurity

TAKE-HOME MESSAGES

❖ LOCAL
PREDATION RISK
INDUCES
ANXIETY-LIKE
STATE

Flexibility



+ Water (Control)

OR

+ Predation signal 1/400

OR

+ Predation signal 1/50

4 populations
60 ind/pop/treatment

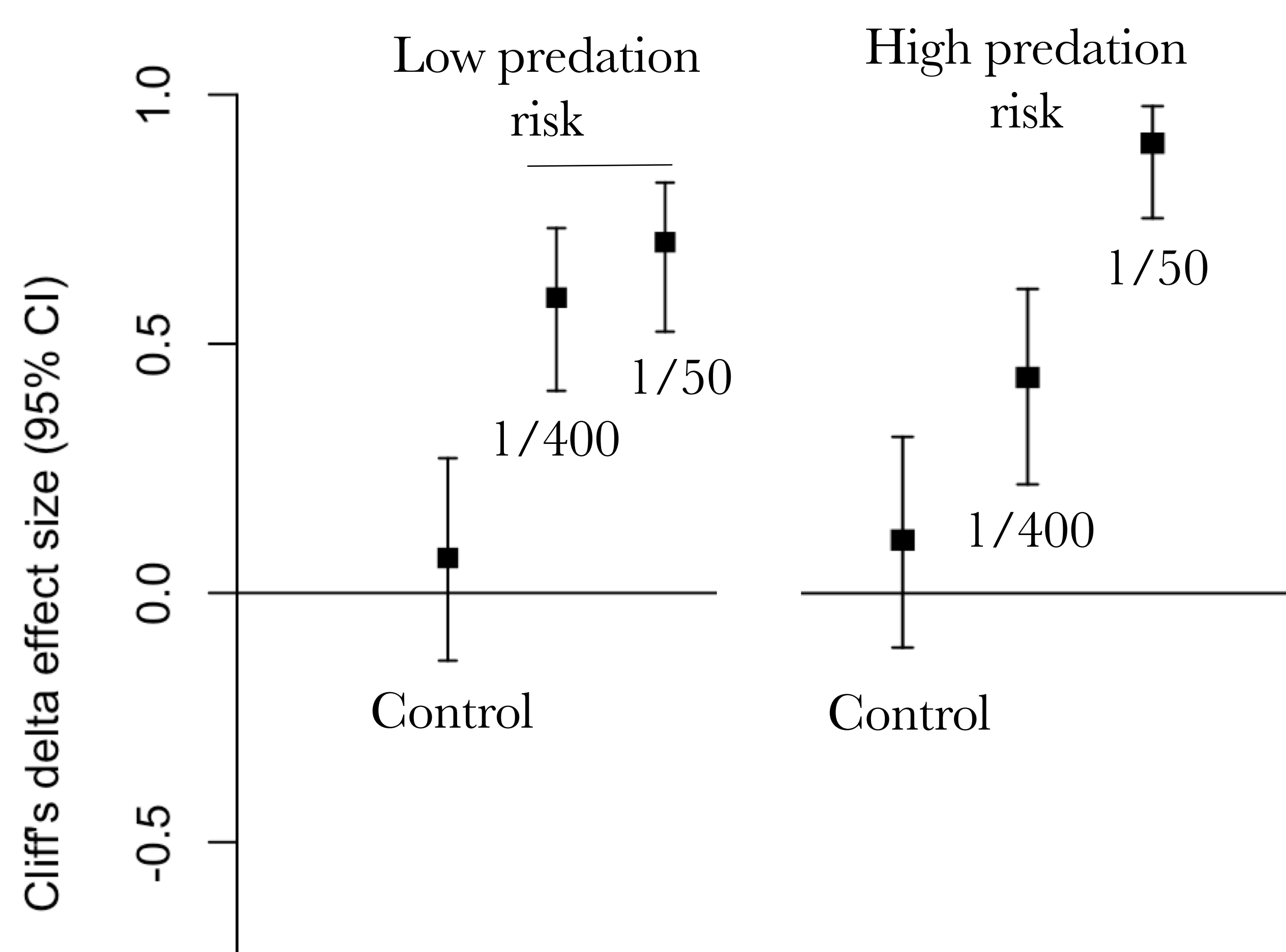


Fig 2. Cliff's delta effect sizes of changes in refuge use after addition of a signal (control or predation signals).

❖ FLEXIBILITY OF
ANTIPREDATOR
RESPONSE

❖ PATTERNS OF
RESPONSES ARE
MODULATED BY
LOCAL PREDATION
RISK